

# Suite Hard · miss review

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Modules 01–06 · 41 items she missed · untimed

Each page is the official item she already sat.

The bar says Module X · #Y so you can match her old sheet.

Module 01 #20 is omitted (she landed).

Module 07 is omitted (no score sheet in the log).

No answers in this file.

Question ID: 1e11190a

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

Store A sells raspberries for **\$5.50** per pint and blackberries for **\$3.00** per pint. Store B sells raspberries for **\$6.50** per pint and blackberries for **\$8.00** per pint. A certain purchase of raspberries and blackberries would cost **\$37.00** at Store A or **\$66.00** at Store B. How many pints of blackberries are in this purchase?

Answer

- A. 4
- B. 5
- C. 8
- D. 12

Question ID: 9f2ecade

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

Question

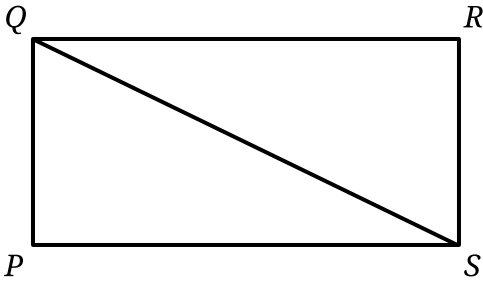
$h(x) = x^3 + ax^2 + bx + c$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are  $-5$ , 6, and 7, what is the value of c ?

Question ID: 2c3aefc9

| Assessment | Test | Domain                    | Skill                            | Difficulty                               |
|------------|------|---------------------------|----------------------------------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Right triangles and trigonometry | Hard <div></div> <div></div> <div></div> |

Question



Note: Figure not drawn to scale.

In rectangle  $PQRS$ , the length of line segment  $QS$  is  $\sqrt{2,344}$  and the length of side  $PS$  is 8 more than the length of side  $RS$ . What is the length of side  $QR$ ?

Question ID: ac472881

| Assessment | Test | Domain  | Skill                            | Difficulty                               |
|------------|------|---------|----------------------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard <div></div> <div></div> <div></div> |

Question

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, *s* and *r* are constants, and *s* > 0. If the equation has infinitely many solutions, what is the value of *s*?

Question ID: 668f1863

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

**Question**  
Function  $f$  is a quadratic function where  $f(-20) = 0$  and  $f(-4) = 0$ . The graph of  $y = f(x)$  in the  $xy$ -plane has a vertex at  $(r, -64)$ . What is the value of  $r$ ?

Question ID: b988eeec

| Assessment | Test | Domain  | Skill            | Difficulty                               |
|------------|------|---------|------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard <div></div> <div></div> <div></div> |

**Question**

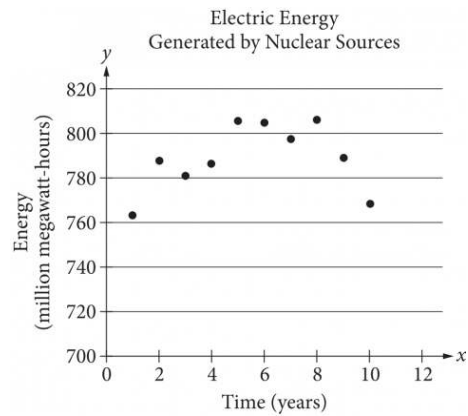
The functions  $f$  and  $g$  are defined as  $f(x) = \frac{1}{4}x - 9$  and  $g(x) = \frac{3}{4}x + 21$ . If the function  $h$  is defined as  $h(x) = f(x) + g(x)$ , what is the x-coordinate of the x-intercept of the graph of  $y = h(x)$  in the xy-plane?

## Question ID: e821a26d

| Assessment | Test | Domain                            | Skill                                      | Difficulty |
|------------|------|-----------------------------------|--------------------------------------------|------------|
| SAT        | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard       |

## Question

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



Of the following equations, which best models the data in the scatterplot?

## Answer

- A.  $y = 1.674x^2 + 19.76x - 745.73$
- B.  $y = -1.674x^2 - 19.76x - 745.73$
- C.  $y = 1.674x^2 + 19.76x + 745.73$
- D.  $y = -1.674x^2 + 19.76x + 745.73$

Question ID: de91a6fc

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

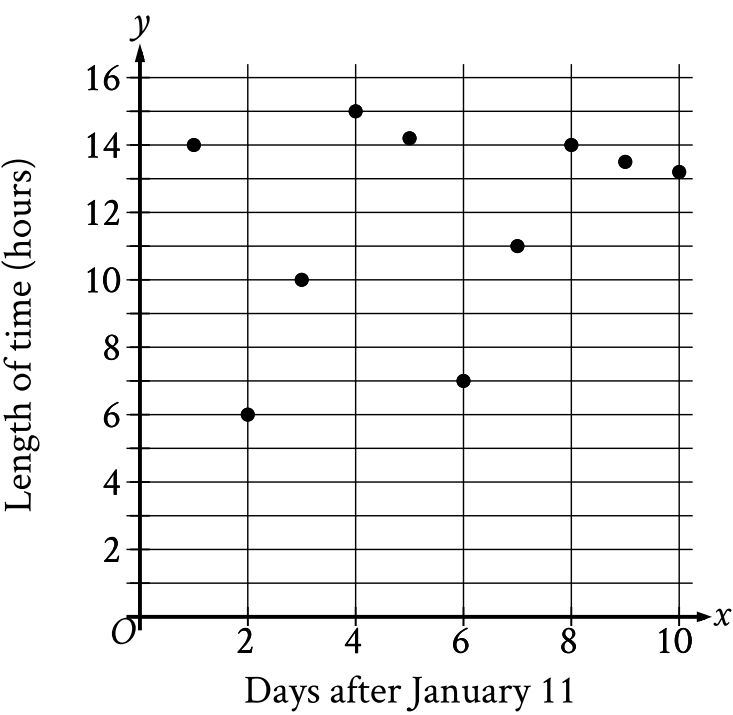
Question

A certain bowl was sold online for prices varying between **\$7** and **\$13**. The number of bowls sold can be modeled by a quadratic function based on the price, where **5** bowls were sold for **\$12** each and the maximum of **9** bowls were sold for **\$10** each. According to the model, what is the number of bowls that were sold for **\$8** each?

Question ID: 7b52985c

| Assessment | Test | Domain                            | Skill                                      | Difficulty                               |
|------------|------|-----------------------------------|--------------------------------------------|------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard <div></div> <div></div> <div></div> |

**Question**  
The scatterplot shows the relationship between the length of time  $y$ , in hours, a certain bird spent in flight and the number of days after January 11,  $x$ .



What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15?

## Question ID: aa43b41f

| Assessment | Test | Domain                            | Skill                                                                | Difficulty    |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard<br>■■■■■ |

## Question

Near the end of a US cable news show, the host invited viewers to respond to a poll on the show's website that asked, "Do you support the new federal policy discussed during the show?" At the end of the show, the host reported that 28% responded "Yes," and 70% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of the population of the United States?

## Answer

- A. The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B. Those who responded to the poll were not a random sample of the population of the United States.
- C. There were not 50% "Yes" responses and 50% "No" responses.
- D. The show did not allow viewers enough time to respond to the poll.

Question ID: af2ba762

| Assessment | Test | Domain  | Skill            | Difficulty                                           |
|------------|------|---------|------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

According to data provided by the US Department of Energy, the average price per gallon of regular gasoline in the United States from September 1, 2014, to December 1, 2014, is modeled by the function  $F$  defined below, where  $F(x)$  is the average price per gallon  $x$  months after September

1.

$$F(x) = 2.74 - 0.19(x - 3)$$

The constant 2.74 in this function estimates which of the following?

**Answer**

- A. The average monthly decrease in the price per gallon
- B. The difference in the average price per gallon from September 1, 2014, to December 1, 2014
- C. The average price per gallon on September 1, 2014
- D. The average price per gallon on December 1, 2014

## Question ID: 7d68096f

| Assessment | Test | Domain                            | Skill                                                                | Difficulty    |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard<br>■■■■■ |

## Question

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week. For the study, the organizer selected **55** teams at random from all trivia teams in a certain tournament. The table displays the information for the **40** teams in the sample that practiced for at least **3** hours per week.

| Hours practiced   | Number of points per round |                   |       |
|-------------------|----------------------------|-------------------|-------|
|                   | 6 to 13 points             | 14 or more points | Total |
| 3 to 5 hours      | 6                          | 4                 | 10    |
| More than 5 hours | 4                          | 26                | 30    |
| Total             | 10                         | 30                | 40    |

Which of the following is the largest population to which the results of the study can be generalized?

## Answer

- A. All trivia teams in the tournament that scored **14** or more points in the round
- B. The **55** trivia teams in the sample
- C. The **40** trivia teams in the sample that practiced for at least **3** hours per week
- D. All trivia teams in the tournament

| Assessment | Test | Domain        | Skill                  | Difficulty                   |
|------------|------|---------------|------------------------|------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard<br>████████████████████ |

The expression  $\frac{1}{3}x^2 - 2$  can be rewritten as  $\frac{1}{3}(x - k)(x + k)$ , where  $k$  is a positive constant. What is the value of  $k$ ?

A. 2

B. 6

C.  $\sqrt{2}$

D.  $\sqrt{6}$

Question ID: 5acbd30

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

**Question**

The function  $f$  is defined by  $f(x) = 56(0.19)^x$ . For any positive integer  $n$ , the value of  $f(n)$  is  $p\%$  less than the value of  $f(n - 1)$ . What is the value of  $p$ ?

- Answer**
- A. 19

B. 44

C. 56

D. 81

## Question ID: 61f61789

| Assessment | Test | Domain                            | Skill                                                | Difficulty    |
|------------|------|-----------------------------------|------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard<br>■■■■■ |

## Question

To study the moisture content in a group of trees, samples from the trunk of each tree were taken from **25** trees and cut in the shape of a cube. The length of the edge of one of these cubes is **2.00** centimeters. If this cube has a mass of **2.56** grams, what is the density of this cube, in grams per cubic centimeter?

Question ID: fb58c0db

| Assessment | Test | Domain                    | Skill   | Difficulty                                        |
|------------|------|---------------------------|---------|---------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div><div></div><div></div><div></div></div> |

Question

Points A and B lie on a circle with radius 1, and arc  $\overline{AB}$  has length  $\frac{\pi}{3}$ . What fraction of the circumference of the circle is the length of arc  $\overline{AB}$ ?

**Question ID: 308084c5**

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                           |
|------------|------|-----------------------------------|------------------------------------------------------|------------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard<br><div><div></div><div></div><div></div></div> |

### Question

| Sample | Percent in favor | Margin of error |
|--------|------------------|-----------------|
| A      | 52%              | 4.2%            |
| B      | 48%              | 1.6%            |

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

### Answer

- A. Sample A had a smaller number of votes that could not be recorded.
- B. Sample A had a higher percent of favorable responses.
- C. Sample A had a larger sample size.
- D. Sample A had a smaller sample size.

| Assessment | Test | Domain  | Skill                             | Difficulty                                           |
|------------|------|---------|-----------------------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | Hard<br><div><div></div><div></div><div></div></div> |

- A. 0
- B.  $-\frac{1}{2}$
- C. -2
- D. The slope of line  $\ell$  is undefined.

**Question ID: 2c5c22d0**

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                   |
|------------|------|---------------|-------------------------------------------------------------------------------|------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard<br>████████████████████ |

### Question

$$y = x^2 + 3x - 7$$

$$y - 5x + 8 = 0$$

How many solutions are there to the system of equations above?

**Answer**

- A. There are exactly 4 solutions.
- B. There are exactly 2 solutions.
- C. There is exactly 1 solution.
- D. There are no solutions.

Question ID: aee9fd2d

| Assessment | Test | Domain  | Skill                            | Difficulty                                           |
|------------|------|---------|----------------------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard<br><div><div></div><div></div><div></div></div> |

### Question

If  $\frac{x+6}{3} = \frac{x+6}{13}$ , the value of  $x + 6$  is between which of the following pairs of values?

**Answer**

- A.  $-7$  and  $-3$   
B.  $-2$  and  $2$   
C.  $2$  and  $7$   
D.  $8$  and  $13$

Question ID: ea6d05bb

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

**Question**  
The expression  $(3x - 23)(19x + 6)$  is equivalent to the expression  $ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants. What is the value of  $b$ ?

**Question ID: db88a933**

| Assessment | Test | Domain                            | Skill                                                              | Difficulty                                           |
|------------|------|-----------------------------------|--------------------------------------------------------------------|------------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard<br><div><div></div><div></div><div></div></div> |

### Question

## Data Set Q

| Value | Frequency |
|-------|-----------|
| $a$   | 110       |
| $b$   | 30        |
| $c$   | 0         |

## Data Set R

| Value | Frequency |
|-------|-----------|
| $a$   | 0         |
| $b$   | 3         |
| $c$   | 11        |

Data sets Q and R are summarized in the tables shown, where  $a$ ,  $b$ , and  $c$  are positive, consecutive integers and  $a < b < c$ . Which statement comparing the means of the data sets is true?

### Answer

- A. The mean of data set R is  $\frac{2(11)}{14}$  less than the mean of data set Q.
- B. The mean of data set R is  $\frac{2(11)}{14}$  greater than the mean of data set Q.
- C. The mean of data set Q is **10** times the mean of data set R.
- D. The means of the two data sets are equal.

## Question ID: 9fdc4eaa

| Assessment | Test | Domain                    | Skill           | Difficulty    |
|------------|------|---------------------------|-----------------|---------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard<br>■■■■■ |

## Question

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

## Answer

- A. **56**
- B. **90**
- C. **106**
- D. **180**

Question ID: 2289b199

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

Question

$$\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[5]{p^8})}$$

For all positive values of  $n$  and  $p$ , the given expression is equivalent to which of the following?

I.  $\frac{\left(n^{\frac{14}{3}}\right)\left(p^{\frac{5}{3}}\right)}{5np}$

II.  $\frac{\sqrt[3]{n^{11}p^2}}{5}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

**Question ID: a7e2859a**

| Assessment | Test | Domain  | Skill            | Difficulty                                           |
|------------|------|---------|------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

The cost of renting a large canopy tent for up to **10** days is **\$430** for the first day and **\$215** for each additional day. Which of the following equations gives the cost ***y***, in dollars, of renting the tent for ***x*** days, where ***x*** is a positive integer and  **$x \leq 10$** ?

**Answer**

- A.  $y = 215x + 215$   
B.  $y = 430x - 215$   
C.  $y = 430x + 215$   
D.  $y = 215x + 430$

Question ID: 43926bd9

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

Question

|     |        |
|-----|--------|
| $x$ | $f(x)$ |
| 1   | $a$    |
| 2   | $a^5$  |
| 3   | $a^9$  |

For the exponential function  $f$ , the table above shows several values of  $x$  and their corresponding values of  $f(x)$ , where  $a$  is a constant greater than 1. If  $k$  is a constant and  $f(k) = a^{29}$ , what is the value of  $k$  ?

Question ID: c8db0e19

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                               |
|------------|------|---------------|-------------------------------------------------------------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div></div> <div></div> <div></div> |

Question

$\frac{1}{4xy} + xyz = \frac{1}{3yz}$

In the given equation,  $x$ ,  $y$ , and  $z$  are positive numbers. Which expression is equivalent to  $y$ ?

Answer

- A.  $\frac{4x-3z}{12x^2z^2}$
- B.  $\sqrt{\frac{4x-3z}{12x^2z^2}}$
- C.  $\frac{1}{3xz^2-4x^2z}$
- D.  $\sqrt{\frac{1}{3xz^2-4x^2z}}$

Question ID: f5aa5040

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                               |
|------------|------|---------------|-------------------------------------------------------------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div></div> <div></div> <div></div> |

**Question**

In the  $xy$ -plane, a line with equation  $2y = c$  for some constant  $c$  intersects a parabola at exactly one point. If the parabola has equation  $y = -2x^2 + 9x$ , what is the value of  $c$ ?

## Question ID: b695711c

| Assessment | Test | Domain                            | Skill                                                                | Difficulty    |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard<br>■■■■■ |

## Question

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of **85** piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for **75** piano students in this sample who practiced at least **50** minutes on average each week.

|                                               | Percentage of correct notes the students played during a recital |            |                  |       |
|-----------------------------------------------|------------------------------------------------------------------|------------|------------------|-------|
| Average number of minutes practiced each week | Less than 50%                                                    | 50% to 80% | Greater than 80% | Total |
| 50 to 150                                     | 7                                                                | 28         | 5                | 40    |
| 151 or more                                   | 3                                                                | 11         | 21               | 35    |
| Total                                         | 10                                                               | 39         | 26               | 75    |

Which of the following is the largest population to which the results of the study can be generalized?

## Answer

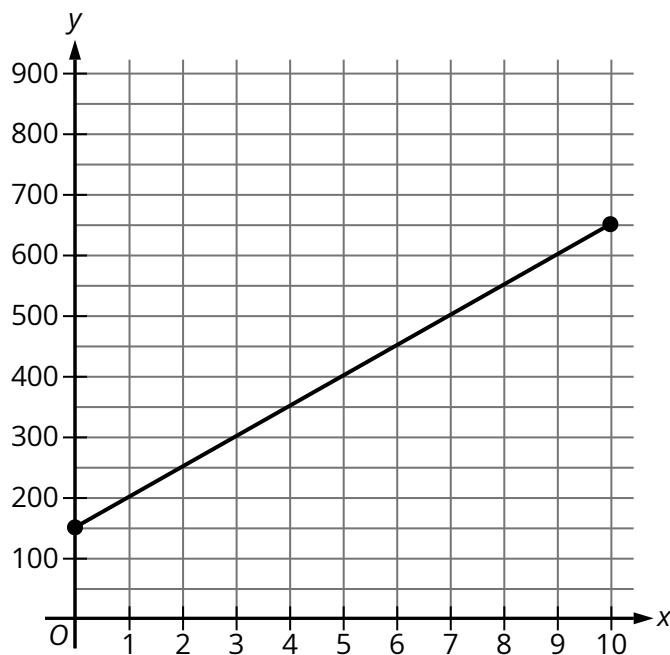
- A. All piano students who were in their second year with their conservatory
- B. All piano students in the region
- C. All piano students in the region who were in their second year with their conservatory
- D. The **75** piano students in the sample who practiced at least **50** minutes on average each week

## Question ID: 590d662d

| Assessment | Test | Domain  | Skill            | Difficulty |
|------------|------|---------|------------------|------------|
| SAT        | Math | Algebra | Linear functions | Hard       |

## Question

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance  $y$ , in meters, Mary had walked on the trail  $x$  minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the **10** minutes after leaving the trail sign?

Question ID: 00165291

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

**Question**

The expression  $6x^4 + 31x^2 + 35$  can be rewritten as  $(3x^2 + a)(2x^2 + b)$ , where  $a$  and  $b$  are positive integers, or as  $(3x^2 + c)(2x^2 + d)$ , where  $c$  and  $d$  are positive nonintegers. What is the value of  $a + c$ ?

**Question ID: 4c5ea142**

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                           |
|------------|------|-----------------------------------|------------------------------------------------------|------------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard<br><div><div></div><div></div><div></div></div> |

### Question

On average, a certain plant grows **87** millimeters every ***m*** months. At this rate, which expression represents the number of millimeters, on average, the plant grows every ***k*** years?

**Answer**

- A.  $\frac{29m}{4k}$   
B.  $\frac{29k}{4m}$   
C.  $\frac{1,044m}{k}$   
D.  $\frac{1,044k}{m}$

Question ID: 5355c0ef

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

Question

$0.36x^2 + 0.63x + 1.17$

The given expression can be rewritten as  $a(4x^2 + 7x + 13)$ , where  $a$  is a constant. What is the value of  $a$ ?

Question ID: 954943a4

| Assessment | Test | Domain                            | Skill       | Difficulty                               |
|------------|------|-----------------------------------|-------------|------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Percentages | Hard <div></div> <div></div> <div></div> |

Question

Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is  $\frac{3}{4}$  cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If  $p$  percent of an adult’s daily allowance of potassium is provided by  $x$  servings of Crunchy Grain cereal per day, which of the following expresses  $p$  in terms of  $x$  ?

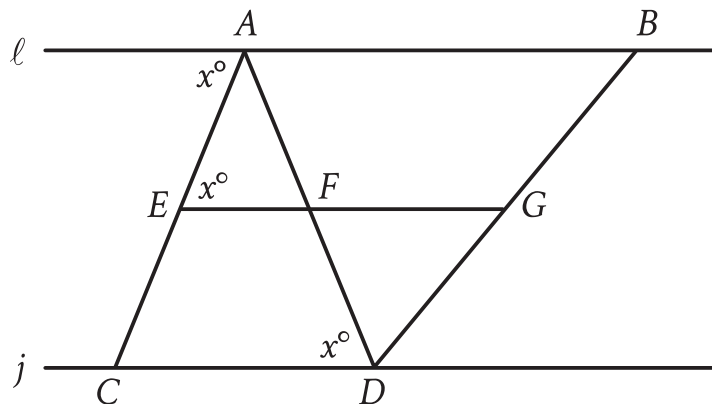
Answer

- A.  $p = 0.5x$
- B.  $p = 5x$
- C.  $p = (0.05)^x$
- D.  $p = (1.05)^x$

Question ID: 858809e7

| Assessment | Test | Domain                    | Skill                        | Difficulty |
|------------|------|---------------------------|------------------------------|------------|
| SAT        | Math | Geometry and Trigonometry | Lines, angles, and triangles | Hard       |

Question



Note: Figure not drawn to scale.

In the figure,  $\overline{EG}$  and  $\overline{AD}$  intersect at point  $F$ , and line  $\overline{EG}$  is parallel to line  $j$ . If  $EF = 0.5CD$  and  $FG = 2.25CD$ , which of the following must be true?

- I.  $AB = 2.25EF$
- II.  $FG = 0.5AB$
- III.  $\frac{EF}{FG} = \frac{AF}{FD}$

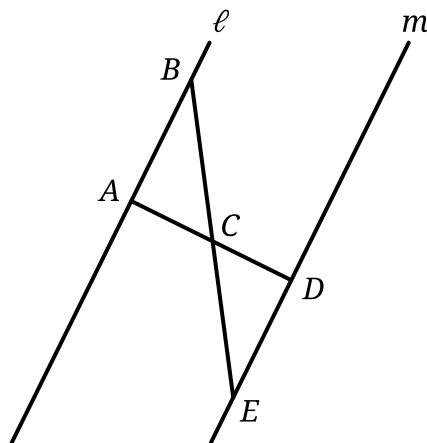
Answer

- A. I only
- B. II only
- C. I and II
- D. II and III

Question ID: 7a11ceea

| Assessment | Test | Domain                    | Skill                        | Difficulty |
|------------|------|---------------------------|------------------------------|------------|
| SAT        | Math | Geometry and Trigonometry | Lines, angles, and triangles | Hard       |

Question



Note: Figure not drawn to scale.

In the figure, segment  $AD$  and segment  $BE$  intersect at point  $C$ . Segment  $AD$  is perpendicular to line  $l$  at point  $A$ . Which of the following additional pieces of information is(are) sufficient to determine that triangle  $ABC$  is congruent to triangle  $DEC$ ?

- I. Segment  $AD$  is perpendicular to line  $m$  at point  $D$ .
- II.  $AC = 13$  and  $DC = 13$

Answer

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

## Question ID: 85939da5

| Assessment | Test | Domain                            | Skill                                                | Difficulty    |
|------------|------|-----------------------------------|------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard<br>■■■■■ |

## Question

| Texting behavior | Talks on cell phone daily | Does not talk on cell phone daily | Total      |
|------------------|---------------------------|-----------------------------------|------------|
| Light            | 110                       | 146                               | 256        |
| Medium           | 139                       | 164                               | 303        |
| Heavy            | 166                       | 74                                | 240        |
| <b>Total</b>     | <b>415</b>                | <b>384</b>                        | <b>799</b> |

In a study of cell phone use, 799 randomly selected US teens were asked how often they talked on a cell phone and about their texting behavior. The data are summarized in the table above. Based on the data from the study, an estimate of the percent of US teens who are heavy texters is 30% and the associated margin of error is 3%. Which of the following is a correct statement based on the given margin of error?

## Answer

- A. Approximately 3% of the teens in the study who are classified as heavy texters are not really heavy texters.
- B. It is not possible that the percent of all US teens who are heavy texters is less than 27%.
- C. The percent of all US teens who are heavy texters is 33%.
- D. It is doubtful that the percent of all US teens who are heavy texters is 35%.

Question ID: 35d37640

| Assessment | Test | Domain                    | Skill   | Difficulty                               |
|------------|------|---------------------------|---------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div></div> <div></div> <div></div> |

**Question**

Point  $F$  lies on a unit circle in the  $xy$ -plane and has coordinates  $(1, 0)$ . Point  $G$  is the center of the circle and has coordinates  $(0, 0)$ . Point  $H$  also lies on the circle and has coordinates  $(-1, y)$ , where  $y$  is a constant. Which of the following could be the positive measure of angle  $FGH$ , in radians?

- Answer**
- A.  $\frac{27\pi}{2}$
  - B.  $\frac{29\pi}{2}$
  - C.  $24\pi$
  - D.  $25\pi$

## Question ID: da978ee6

| Assessment | Test | Domain                            | Skill                                                | Difficulty    |
|------------|------|-----------------------------------|------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard<br>■■■■■ |

## Question

A researcher surveyed a random sample of **1,020** people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that **86%** of people from the city support this proposal, with an associated margin of error of **2.13%**. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

## Answer

- A. The margin of error is between **2.13%** and **3.13%**.
- B. The margin of error is between **4.13%** and **5.13%**.
- C. The margin of error is greater than **5.13%**.
- D. The margin of error is less than **2.13%**.

Question ID: 429fb7c0

| Assessment | Test | Domain  | Skill                            | Difficulty                                        |
|------------|------|---------|----------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard <div><div></div><div></div><div></div></div> |

**Question**  
What value of  $t$  is the solution to the equation  $0.8t - 0.46 = 8(t - 0.001) + 1.9$ ?

**Question ID: 58dcc59f**

| Assessment | Test | Domain        | Skill               | Difficulty    |
|------------|------|---------------|---------------------|---------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard<br>■■■■■ |

**Question**

A landscaper is designing a rectangular garden. The length of the garden is to be 5 feet longer than the width. If the area of the garden will be 104 square feet, what will be the length, in feet, of the garden?